

ICFIRE 2026

INTERNATIONAL CONFERENCE ON FIRE SAFETY
AND RESILIENCE OF THE BUILT ENVIRONMENT



PROCEEDINGS

22-23 JULY 2026





Amber Laboratory (Pvt) Ltd.

ICFIRE 2026

International Conference on Fire Safety and Resilience of the Built Environment

Advancing Fire-Resilient Materials and Systems for
Future-Proofing Sustainable Built Environments

Abstracts

22nd - 23rd July 2026

This book contains the abstracts of the presentations delivered at ICFIRE-2026, the International Conference on Fire Safety and Resilience of the Built Environment, jointly organized by Amber Laboratory (Pvt) Ltd, Sri Lanka, the Society of Fire Protection Engineers (SFPE) Sri Lanka Chapter, and the University of Moratuwa. The conference was held on 22nd - 23rd July 2026.

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International Conference on Fire Safety and Resilience of the Built Environment
Theme - “Advancing Fire-Resilient materials and systems for future-proofing sustainable Built Environments”
Amber Laboratory Pvt Ltd.
22 to 23rd July 2026
Lotus Hall, BMICH, Colombo, Sri Lanka

Inaugural Program

Day 1 - Wednesday, July 22nd, 2026	
8.00 a.m	Arrival of Guests and Registration
9.00 a.m	National Anthem and Lighting of the Traditional Oil Lamp
9.10 a.m	Welcome Address Dr. Radhika Samarasekara, Executive Director, Amber Laboratory (Pvt) Ltd., Sri Lanka
9.15 a.m	Introductory Address and Corporate Video Amber Laboratory (Pvt) Ltd., Sri Lanka
9.25 a.m	Address by the Chief Guest Hon. Prof. Chrishantha Abeysena, Minister of Science & Technology of Sri Lanka
9.40 a.m	Address by the Guest of Honor Prof. Priyan Mendis, School of Electrical, Mechanical and Infrastructure Engineering, University of Melbourne, Australia
9.50 a.m	Address by the Special Guest Dr. Ryan Hilditch, Principal, Halliwell Fire Science, Australia and New Zealand, Australia
10.00 a.m	Address by the Special Guest and Platinum Sponsor Eng. Vajira Attanayake, Group Director, Airow Solutions (Pvt) Ltd., Sri Lanka
10.10 a.m	Keynote Address Fire Engineering and Design: Emerging Trends in Performance Testing and Compliance Prof. Priyan Mendis, School of Electrical, Mechanical and Infrastructure Engineering, University of Melbourne, Australia
10.50 a.m	Vote of Thanks & Group Photo
11.05 a.m	COFFEE BREAK

International Conference on Fire Safety and Resilience of the Built Environment
Technical Program

Day 1 - Wednesday, July 22nd, 2026	
Technical Session 1	
11.20 a.m - 12.00 p.m	Holistic Fire Safety Engineering: From Fundamentals to Real-World Applications Dr. Ryan Hilditch, Principal, Halliwell Fire Science, Australia and New Zealand
12.00 p.m - 12.30 p.m	Fire Protection and Resilience in Historical and Heritage Buildings Architect, Ashley de Vos, Consultant, ADV Consultants, Sri Lanka
12.30 p.m - 1.30 p.m	LUNCH
Technical Session 2	
1.30 p.m - 2.00 p.m	Research and Innovation in Fire Research from Materials to Artificial Intelligence (AI) Prof. Kate Nguyen, Leader of Fire Engineering & Deputy Head of Department of Civil & Infrastructure Engineering, RMIT University, Melbourne, Australia
2.00 p.m – 2.30 p.m	Modern Built Environments: Architectural Design Perspective to meet Fire Safety Requirements Architect, Athula Amarasekera, Founder and CEO, Design Team 3 Pvt Ltd., Singapore
2.30 p.m – 3.00 p.m	Structural Fire Engineering and Performance-Based Design in the Built Environment Dr. Imran Ahamed, Senior Fire Consultant, Halliwell Fire, Brisbane, Australia
3.00 p.m. 3.30 p.m	COFFEE BREAK
Technical Session 3	
3.30 p.m – 4.00 p.m	The Fire Regulations, Codes, and Enforcement Frameworks in Sri Lanka Eng. Kosala Kamburadeniya, President of the Institute of Engineering Sri Lanka and Managing Director and Principal Consultant, K2 Consultants (Pvt) Ltd, Sri Lanka

4.00 p.m – 4.40 p.m	Recent Advances in Modelling Techniques for Fire Safety and Resilience in Key Buildings Eng. Dilan Kahatapitiya, Manager of Training and Research, Civil and Structural Engineering Consultancy (CSEC) Pvt Ltd., Sri Lanka
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Day 2 Thursday, July 23rd, 2026	
Technical Session 4	
9.00 a.m – 9.30 a.m	Emerging Fire Risks in Urban Infrastructure: Battery Energy Storage Systems (BESS) and Electric Vehicles (EV) Dr. Ryan Hilditch, Principal, Halliwell Fire Science, Australia and New Zealand
9.30 a.m – 10.00 a.m	Modern Built environments: Structural Design Perspective to meet Fire Safety Requirements Eng. Shiromal Fernando, Founder & Managing Director, Civil and Structural Engineering Consultancy (CSEC) Pvt Ltd., Sri Lanka
10.00 a.m – 10.40 a.m	Applications and Performance of Fire-resistant Materials Eng. Vajira Attanayake, Group Director, Airow Solutions (Pvt) Ltd., Sri Lanka
10.40 a.m – 11.00 am	COFFEE BREAK
Technical Session 5	
11.00 a.m – 11.30 a.m	Understanding Fire Dynamics, Fire Growth, Spread, and Smoke Movement Dr Pasindu Weerasinghe, Dept of Civil Engineering, University of Moratuwa, Sri Lanka
11.30 a.m- 12.00 a.m	Fire Investigation in Practice: Case Studies, Lessons Learned and Pathways to Reform Mrs Vithyaa Thavapalan, Founder, Managing Director and Forensic Fire Investigator, Forensic Origin and Cause Investigations, Australia
12.00 p.m -12.40 p.m	From Laboratory Testing to Compliance: Assessment of Passive Fire Protection Products Dr. Imran Ahamed, Senior Fire Consultant, Halliwell Fire, Brisbane, Australia
12.40 p.m –1.40 p.m	LUNCH

Technical Session 6	
1.40 p.m – 2.10 p.m	Forensic Fire Investigation: Reconstruction of Fire Using Fire Engineering Tools Dr Aatif Ali Khan, Senior Lecturer, Civil and Environmental Engineering, University of Canterbury, New Zealand
2.10 p.m – 2.40 p.m	Fire Safety Challenges and Innovative Façade Solutions in Buildings with Glass Envelopes Dr.Uthpala Rathnayake, Senior Lecturer, Dept of Facility Management, Faculty of Architecture, University of Moratuwa
2.40 p.m – 3.10 p.m	AI-Driven IoT-based fire protection for the Manufacturing Industry Dr Prabath Weerasinghe, Dept of Electrical and Information Engineering, Faculty of Engineering, University of Ruhuna
3.10 p.m – 3.30 p.m	COFFEE BREAK
3.30 p.m – 4.45 p.m	PRACTIONER'S FORUM - Developing the Colombo Declaration
4.45 p.m – 5.00 p.m	CLOSING CEREMONY
END OF THE CONFERENCE	

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Message from the Honorable Minister of Science and Technology



It is with great pleasure that I extend my warm greetings to the organizers, keynote and plenary speakers, researchers, industry and academic professionals, policymakers, and all participants of the International Conference on Fire Safety and Resilience of the Built Environment (ICFIRE 2026).

I commend Amber Laboratory, the Society of Fire Protection Engineers (SFPE) Sri Lanka Chapter, and the University of Moratuwa for organizing this important international conference, which provides a valuable platform for sharing knowledge, fostering innovation, and strengthening collaboration in the field of fire safety and resilience.

The theme of ICFIRE 2026, “Advancing Fire-Resilient Materials and Systems for Future-Proofing Sustainable Built Environments,” is both timely and highly relevant. It reflects the need for multidisciplinary collaboration in developing innovative materials, advanced engineering solutions, performance-based design approaches, and evidence-based policies that will enhance the resilience of the built environment.

I am confident that the deliberations and outcomes of ICFIRE 2026 will contribute significantly to strengthening fire safety practices in Sri Lanka and across the region and to advancing research excellence, innovation, policy development, regulatory strengthening, and knowledge dissemination.

The Ministry of Science and Technology remains committed to promoting scientific research, technological innovation, and strong partnerships among government, academia, research institutions, professional organizations, and industry. Such collaborations are essential for translating research into practical solutions that improve public safety, strengthen national resilience, and contribute to socio-economic development.

I congratulate the organizers for their vision, commitment, and dedication in bringing together an outstanding forum of national and international experts to address one of the most critical aspects of sustainable development and public safety. I extend my best wishes for a productive and successful conference.

Hon. Prof. Chrishantha Abeysena (M.P.)
Minister of Science & Technology
Democratic Socialist Republic of Sri Lanka

Message from the Guest of Honor



It is a privilege to contribute this message to the proceedings of the International Conference on Fire Safety and Resilience of the Built Environment (ICFIRE 2026).

At a time when our built environment is being increasingly challenged by rapid urbanization, taller and more complex buildings, complex construction methods, and evolving architectural designs, this conference provides an essential platform to advance knowledge, innovation, and collaboration in fire safety and resilience.

Traditional prescriptive approaches to fire protection are increasingly being complemented. Today, performance-based fire engineering enables designers and engineers to evaluate complex fire scenarios, optimize safety strategies, and achieve regulatory compliance through scientific analysis and engineering judgment. When these elements are aligned, the consequences of fire are dramatically mitigated. Conversely, fragmented approaches create vulnerabilities that place lives, property, and critical infrastructure at risk.

This conference breaks down silos, enabling the translation of research into practical standards, regulations, and engineering practice. Equally important is the systematic collection and sharing of high-quality data on fire incidents and building performance to support evidence-based policymaking, improve regulatory frameworks, and optimize resource allocation.

I encourage all participants to use this conference not only as a forum for exchanging ideas but also to form partnerships, joint research, and policy initiatives. Governments, academia, and industry must also commit to funding applied research and capacity building so that innovations reach the communities that need them most.

In closing, I commend all contributors to ICFIRE 2026 for their dedication to advancing the science and practice of fire safety and resilience. I also wish to express my appreciation to Amber Laboratory, the Society of Fire Protection Engineers (SFPE) Sri Lanka Chapter, and the University of Moratuwa for their leadership and commitment in organizing ICFIRE 2026. I extend my best wishes for a successful conference

Prof. Priyan Mendis

Professor, School of Electrical, Mechanical, and Infrastructure Engineering, University of Melbourne, Australia

Message from the Special Guest



It is a great honor to be invited to speak at this conference and to join such a distinguished gathering of engineers, researchers, fire service professionals, industry representatives, and honored guests. It is a privilege to be here in Sri Lanka, and I am sincerely grateful for the warm welcome. I believe events such as this play an important role in supporting Sri Lanka's future development by bringing together expertise, encouraging innovation, and strengthening the profession. I believe that making international connections and new friends enriches our lives.

As our built environment evolves in response to new technologies, changing climates, and increasing urbanization, collaboration and the exchange of knowledge have never been more important. Conferences such as this provide an invaluable opportunity to learn from one another and to advance our profession together in ways that support safer, stronger, and more resilient communities.

I hope my presentations can encourage discussion around performance-based fire engineering and inspire engineers and colleagues to think beyond compliance towards creating safer, more resilient buildings and societies for future generations.

Dr. Ryan Hilditch

Principal, Halliwell Fire Science, Australia and New Zealand

Message from the Special Guest and the Platinum Sponsor



It is with great pride and pleasure that Airow Solutions Pvt. Ltd. joins ICFIRE 2026 as a Platinum Sponsor. We extend our sincere appreciation to Amber Laboratory for organizing this prestigious international conference, providing an exceptional platform for researchers, academics, industry professionals, and innovators to exchange knowledge and drive advancements in fire engineering and safety.

As the built environment continues to evolve, the importance of resilient infrastructure and advanced fire protection systems has never been greater. Through collaboration between academia, research institutions, regulatory bodies, and industry, we can develop safer, more sustainable solutions that protect lives, assets, and communities. Conferences such as ICFIRE play a vital role in fostering these partnerships and accelerating innovation.

At Airow Solutions, we are committed to delivering world-class engineering solutions across Sri Lanka, the Maldives, and the wider region. As the exclusive agent for Nullifire (UK) and FireSpray (UK), we provide internationally recognized passive fire protection systems alongside our expertise in structural rehabilitation, waterproofing technologies, industrial protective coatings, and specialized engineering solutions for some of the world's most demanding environments. Our mission is to deliver innovative, high-performance solutions that enhance the safety, durability, and sustainability of modern infrastructure.

We firmly believe that continuous learning, research, and knowledge sharing are fundamental to engineering excellence. We are honored to support an event that brings together distinguished experts and emerging professionals to address today's challenges while shaping the future of fire safety engineering.

On behalf of Airow Solutions Pvt. Ltd., we congratulate Amber Laboratory and the organizing committee for their dedication and vision in making ICFIRE 2026 a success. We wish all delegates a productive conference, engaging discussions, and lasting professional connections.

Together, let us build a safer, smarter, and more resilient future through innovation, collaboration, and engineering excellence.

Eng. Vajira Attanayake

Group Director, Airow Solutions Pvt. Ltd, Sri Lanka & Maldives

Message from the President - SFPE Sri Lankan Chapter



On behalf of the Society of Fire Protection Engineers (SFPE) – Sri Lanka Chapter, I congratulate Amber Laboratory on convening the inaugural International Conference on Fire Safety and Resilience of the Built Environment, held on the 22nd and 23rd of July 2026 at the BMICH, Colombo. The theme, “Advancing Fire-Resilient Materials and Systems for Future-Proofing Sustainable Built Environments,” is a timely one. As our cities grow taller and denser, and as new materials, façade systems, battery energy storage, and electric mobility reshape the built environment, the fire risks we must manage are changing. Fire safety is no longer a matter of prescriptive compliance alone; it calls for sound science, performance-based design, and collaboration across disciplines.

This conference spans fire dynamics, structural fire engineering, forensic investigation, heritage buildings, emerging infrastructure risks, and the growing use of modelling and artificial intelligence, bringing together contributors from Sri Lanka, Australia, and New Zealand, drawn from universities, research institutions, and practice. For the SFPE Sri Lanka Chapter, this reflects the work we exist to support: advancing the practice of fire protection engineering and connecting local practitioners, researchers, and regulators with international expertise. I am also glad that the conference looks towards collective action, through the Practitioners’ Forum and the development of the Colombo Declaration.

I thank Amber Laboratory and the organising committee, the speakers, the sponsors, and the delegates whose participation made the event possible. I hope the ideas shared over these two days continue to contribute to a safer and more resilient built environment in Sri Lanka.

With regards

Dr. Pasindu Weerasinghe

President, Society of Fire Protection Engineers (SFPE), Sri Lanka Chapter

Message from the Organizing Chair and Executive Director, Amber Laboratory



It is my great pleasure, on behalf of Amber Laboratory and the organizing committee, to welcome you to the International Conference on Fire Safety and Resilience of the Built Environment (ICFIRE 2026). ICFIRE 2026, held on 22–23 July 2026 at the BMICH in Colombo, Sri Lanka.

We are honoured to welcome Hon. Prof. Chrisantha Abeysena, Minister of Science & Technology of Sri Lanka, as the Chief Guest. We are also privileged to have Prof. Priyan Mendis of the University of Melbourne, Australia, as the Guest of Honour and Keynote Speaker, together with Dr. Ryan Hilditch, Principal, Halliwell Fire Science, Australia and New Zealand, and Eng. Vajira Attanayake, Group Director of Airow Solutions (Pvt) Ltd., Sri Lanka and Maldives, as our Special Guests.

ICFIRE 2026 marks an important milestone in advancing fire safety science, engineering practice, and resilience across the global built environment. Under the theme “Advancing Fire-Resilient Materials and Systems for Future-Proofing Sustainable Built Environments,” the conference provides a premier international platform for engineers, architects, researchers, fire professionals, policymakers, industry leaders, facility managers, urban planners, and policymakers to exchange knowledge, innovations and address emerging fire safety challenges. A key outcome of the conference is the Colombo Declaration on Fire Resilience in the Built Environment (2026), which presents strategic recommendations for strengthening fire resilience at local, national, and international levels.

The conference features keynote and technical presentations, practitioners' forum, and expert discussions including structural fire engineering and design, protection, fire-resistant materials, testing, investigations, and regulations.

I extend my sincere appreciation to our co-organizers, the Society of Fire Protection Engineers (SFPE) Sri Lanka Chapter and the University of Moratuwa, together with our distinguished speakers, local and international scientific committees, organizing committee, partners, and all participants for their invaluable support. Special thanks to our platinum and other sponsors for financial support and the Directors and staff of Amber Laboratory for their dedication in making ICFIRE 2026 a success.

Wish you all a productive, engaging, and memorable conference.

Dr Radhika Samarasekera

Organizing Chair and Executive Director, Amber Laboratory, Sri Lanka

Keynote Address

Fire Engineering and Design: Emerging Trends in Performance Testing and Compliance



Prof. Priyan Mendis

*Professor, School of Electrical, Mechanical and Infrastructure Engineering,
University of Melbourne, Australia*

Abstract

Rapid urbanization, increasing building heights, innovative construction materials, and evolving architectural designs have significantly transformed the challenges faced in modern fire safety engineering. Traditional prescriptive approaches to fire protection are increasingly being complemented and, in many cases, replaced by performance-based fire engineering methodologies that enable designers to evaluate complex fire scenarios, optimize safety strategies, and achieve regulatory compliance through scientific analysis and engineering judgment.

This presentation explores the latest developments in fire engineering and design, focusing on emerging trends in performance-based design, fire testing methodologies, structural fire engineering, digital simulation technologies, and evolving compliance frameworks. It examines how computational fire modelling, smoke movement analysis, evacuation modelling, and structural fire assessments are being integrated into the building design process to enhance occupant safety while supporting architectural innovation and sustainable development.

The presentation also discusses the importance of fire testing methods and recent advancements, including material fire resistance, façade performance testing, compartment fire experiments, and large-scale validation methods that improve the understanding of fire behavior in modern construction systems. Particular attention is given to the fire performance of emerging materials such as mass timber, high-strength steel, ultra-high-performance concrete, composite structural systems, and

battery energy storage installations, which introduce both opportunities and new fire safety challenges.

The presentation further reviews international regulatory developments and compliance frameworks, including performance-based provisions. The discussion highlights the growing importance of engineering-based compliance demonstrations for complex buildings where conventional prescriptive requirements may not adequately address unique design challenges.

Through selected international case studies, including Sri Lanka, the successful application of fire engineering principles in medium and high-rise buildings and infrastructure will be presented. The presentation will also include future directions in fire engineering, emphasizing resilient design, sustainability, digital innovation, and multidisciplinary collaboration as essential components for delivering safer, smarter, and more sustainable built environments.

Plenary Lectures

Holistic Fire Safety Engineering: From Fundamentals to Real-World Applications



Dr. Ryan Hilditch

Principal, Halliwell Fire Science, Australia and New Zealand

Abstract

Most engineers are familiar with designing to the requirements of building codes, where compliance is demonstrated by following prescriptive rules. But what happens when a building is too complex, innovative, or unconventional for those rules to provide a practical solution? This is where performance-based fire engineering offers a different approach. Rather than simply asking whether a design complies with a code, performance-based fire engineering asks whether the building will achieve its fire safety objectives. By applying engineering principles, fire science, and analytical tools, fire safety can be considered as part of a holistic building design, balancing fire performance with architectural vision, structural efficiency, functionality, and constructability. In this talk, we will explore fundamental concepts of performance-based fire engineering, discuss when and why it is used, and outline the principles of developing and accessing a performance-based solution.

Fire Protection and Resilience in Historical and Heritage Buildings



Ashley de Vos

Chartered Architect, Landscape Architect, Conservator, and Heritage Management Consultant, ADV Consultants, Sri Lanka

Abstract

In the Historical Period, Designs were executed by humans for humans, with space and user safety in mind. The buildings opened into a large open space, to courtyards, allowing people to disperse easily. Easy, wide access was based on the number of people using the facility. The materials used were specially selected to be suitable for the task. Nothing was based on experimentation; most respected their limitations and rose above them. While there was no specific interest in fire safety, the limit on building height and the use of solid, tested material helped. The brick buildings of the past are longer-lasting, some 2000 years or more, than the new exhibition pavilion structures being experimented with today. Concrete has a life no one speaks about. Sri Lanka was saved as it did not experience the urbanisation seen in Europe. Unfortunately, the new, hurried copy-paste of the past fifteen years is cutting down trees and robbing the city of much-needed shade. While creating clones to live in artificial environments.

Today, we are arrogant and think we can play god; we think that regulations, mostly borrowed from abroad, if followed, will save us. We are great dreamers, as we also know that the moron called the politician will break all laws to get what they want. Just because we studied abroad and followed the 18th century MacCaully principle that created black and white men, who think they have superior knowledge, and can force their views of destruction in an unrelated urbanisation, forced on by the Bretton Woods twins, who have their own destructive agenda. We are unable to acknowledge the collective genius of the past and depend on technology to save us. All technology has its limitations. Driven by greed, we rise to the occasion by becoming great imitators,

willing to adopt the failures drawn from cultureless countries. We are willing to destroy Sri Lanka for our greed-driven obsession. We are creating clones to live in sterile environments, utilising cheap experimental materials. Planning has been left to the developer, who wants to utilise the site's maximum footprint to increase his profit. Instead, the UDA should produce the planned regulatory building envelopes that should be followed. Regulatory bodies like the UDA have been reduced to rubber stamps that readily yield to the developer's dictates in areas with no infrastructure. Colombo is a great swamp with three hills; much of it was filled in by the British. All development is in this hurriedly filled swamp. How stable is that?

Modern Built Environments: Architectural Design Perspective to meet Fire Safety Requirements



Architect, Athula Amarasekera

Founder and CEO, Design Team 3 Pvt Ltd., Singapore

Abstract

The paper will compare the fire codes of Sri Lanka with other international codes and practices in design in terms of building typologies relevant to Sri Lanka. Also presented will be some of the findings of recent major fires in buildings and how Sri Lanka needs to strengthen its design and regulatory framework to avoid similar events.

Design tools that can be used to enhance fire safety design as well as building efficiency, such as performance-based analysis vs code compliance, will be explored, especially in terms of new building typologies.

Modern Built environments: Structural Design Perspective to meet Fire Safety Requirements



Eng. Shiromal Fernando

*Founder & Managing Director, Civil and Structural Engineering Consultancy (CSEC)
Pvt Ltd., Sri Lanka*

Abstract

Modern built environments are increasingly characterized by high-rise construction, mixed-use developments, large open spaces, complex architectural forms, underground facilities, interconnected buildings, and the extensive use of lightweight and high-strength construction materials. Although these developments provide functional, economic, and architectural benefits, they also introduce significant challenges in maintaining structural stability during a fire. Fire safety must therefore be considered not only as a matter of evacuation, detection, suppression, and regulatory compliance, but also as a fundamental structural design requirement.

This presentation examines the role of structural engineering in achieving adequate fire safety within modern buildings and infrastructure. It discusses how elevated temperatures influence the strength, stiffness, deformation, connection behavior, and load-carrying capacity of structural materials, including reinforced concrete, structural steel, composite systems, masonry, and engineered timber. Particular attention is given to the redistribution of forces, thermal expansion, restraint effects, local and global instability, progressive collapse, and the potential loss of structural robustness under severe fire conditions. The presentation further evaluates the limitations of relying solely on prescriptive fire-resistance ratings, especially for buildings with unusual geometries, large compartments, high occupant densities, complex structural systems, or critical operational functions. While prescriptive requirements provide a practical minimum level of protection, they may not adequately represent the actual fire exposure, structural interaction, load paths, or failure mechanisms of complex developments. Performance-based structural fire engineering is therefore presented

as an important complementary approach. This method enables fire scenarios, thermal actions, structural response, evacuation requirements, firefighting operations, and acceptable levels of damage to be assessed in an integrated and project-specific manner.

Key structural design considerations include appropriate fire compartmentation, continuity and redundancy of load paths, protection of critical structural members and connections, resistance to disproportionate collapse, allowance for thermal movements, and coordination between passive and active fire-protection systems. The importance of early collaboration among structural engineers, architects, fire engineers, building-services engineers, contractors, approving authorities, and emergency-response stakeholders is also emphasized. Such coordination is essential to ensure that architectural objectives, construction methods, service penetrations, fire-protection systems, and structural performance requirements remain compatible throughout design, construction, and operation.

The presentation concludes that fire safety in modern built environments cannot be effectively achieved through isolated systems or code compliance alone. A resilient building must be capable of facilitating safe evacuation, supporting firefighting operations, limiting fire and smoke spread, preventing premature structural collapse, and reducing the potential for cascading failure. Integrating structural fire engineering principles from the earliest stages of design can therefore enhance life safety, structural resilience, post-fire recoverability, and the long-term sustainability of the built environment.

The Fire Regulations, Codes, and Enforcement Frameworks in Sri Lanka



Eng. Kosala Kamburadeniya

President of the Institute of Engineering, Sri Lanka, and Managing Director and Principal Consultant, K2 Consultants (Pvt) Ltd, Sri Lanka

Abstract

Effective fire regulations, codes, and enforcement are essential to reduce risk, save lives, protect property, sustain business continuity, and preserve public health and the environment. This presentation examines the evolution, scope, and practical impact of fire and life-safety regulations in Sri Lanka and situates them in comparison with international standards.

Key themes include:

Objectives of fire regulation: life safety, property and asset protection, business continuity, public health and environmental protection, and clarity for designers and builders.

The regulatory landscape in Sri Lanka: history and development of the CIDA Fire Regulations, relevant Gazette notifications, and other statutory instruments that define legal obligations for developers, contractors, and professionals.

Roles and responsibilities: the Authority Having Jurisdiction (AHJ), Fire Services Department, CIDA, and other institutional stakeholders—their enforcement powers, permitting processes, inspection regimes, and incident-review functions.

Technical and procedural framework: the lifecycle of fire-safety delivery from risk assessment and preliminary design to installation, commissioning, periodic inspection, testing, and maintenance of systems.

Enforcement mechanisms: permitting, trained inspectors, regular inspections, clear penalties, public education, and post-incident review as means to translate standards into real-world protection and continuous improvement.

Comparison with international codes and best practices: identifying gaps relative to global standards in design criteria, performance verification, competency frameworks, and enforcement capacity.

Implementation challenges: institutional capacity constraints, ambiguity in responsibilities, weak enforcement and penalty regimes, fragmented statutory authority, and the pressures of increasingly complex building typologies.

Priority recommendations: immediate actions to strengthen enforcement and compliance enhanced training and accreditation for inspectors and practitioners, clearer AHJ mandates, improved permitting and inspection workflows, mandatory commissioning and maintenance regimes, stronger penalties and incentives, harmonization with international standards, and expanded public awareness and industry engagement.

This presentation argues that a robust, well-resourced enforcement framework supported by clear legal mandates, skilled institutions, consistent technical standards, and active stakeholder collaboration is critical to ensuring fire and life safety in Sri Lanka's rapidly evolving built environment.

Structural Fire Engineering and Performance-Based Design in the Built Environment



Dr. Imran Ahamed

Senior Fire Consultant, Halliwell Fire, Brisbane, Australia

Abstract

Structural fire engineering is the discipline concerned with understanding and predicting how building structures behave when exposed to fire, and with designing structural elements to maintain adequate performance for a defined period during a fire event. This presentation provides an overview of structural fire engineering principles and explores how performance-based design is increasingly used alongside, and at times in place of, traditional prescriptive requirements. The discussion begins with the fundamentals of structural fire engineering, including how elevated temperatures affect material properties and structural capacity. It then outlines prescriptive fire resistance requirements as set out in relevant codes and standards, and how these requirements are typically satisfied through standard construction practices and tested or deemed-to-satisfy solutions. The presentation then examines design approaches for fire resistance across the major structural materials, concrete, steel, timber, and composite construction, highlighting the differing behaviour and protection strategies relevant to each. A key focus is the limitations of prescriptive approaches, particularly where building geometry, material innovation, or project constraints make compliance difficult or inefficient. The presentation discusses how performance-based design offers greater design flexibility, enabling alternative solutions that can achieve equivalent or superior safety outcomes while also delivering cost savings and architectural or commercial value. The presentation concludes with case studies illustrating the practical application of performance-based structural fire design, demonstrating how engineering analysis can be used to justify alternative solutions and support compliance.

Understanding Fire Dynamics, Fire Growth, Spread, and Smoke Movement



Dr. Pasindu Weerasinghe

Department of Civil Engineering, University of Moratuwa, Sri Lanka

Abstract

Fire dynamics provides the scientific foundation on which fire safety engineering rests. A clear understanding of how fires ignite, grow, and spread, and how the smoke they generate moves through a building is essential for the rational design of detection, suppression, evacuation, and smoke management systems. This presentation introduces the principles that govern the behavior of fire in the built environment and connects them to the decisions faced by engineers, architects, and regulators.

The talk begins with the chemistry and physics of combustion and the concept of heat release rate as the single most important descriptor of a fire. It traces the characteristic phases of compartment fire development: ignition, growth, flashover, the fully developed stage, and decay, and examines the conditions that drive the transition to flashover, along with the distinction between fuel-controlled and ventilation-controlled burning. The three modes of heat transfer are discussed as the mechanisms behind flame spread across surfaces and between compartments, including the modern materials and combustible façades on the rate of spread. The second part addresses smoke, a leading cause of injury and death in building fires. It considers smoke production, buoyancy, and the stack effect, and the movement of smoke through corridors, shafts, and stairwells, together with the principles of smoke control and the concept of tenability. They are drawn together through the framework of available versus required safe egress time (ASET/RSET), showing how an understanding of fire dynamics underpins performance-based design and life-safety assessment.

The session aims to provide a shared vocabulary and a physical intuition that practitioners across disciplines can apply when evaluating fire risk and designing safer, more resilient buildings.

Emerging Fire Risks in Urban Infrastructure: Battery Energy Storage Systems (BESS) and Electric Vehicles (EV)



Dr. Ryan Hilditch

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Abstract

Batteries are becoming an essential part of modern life, powering everything from smartphones and electric vehicles to homes, businesses, and large-scale battery energy storage systems (BESS). As the transition to cleaner energy accelerates, electric vehicles and BESS are being adopted at an unprecedented rate. Alongside their many benefits, however, these technologies introduce new and evolving fire safety challenges that differ significantly from those associated with conventional vehicles, traditional fuels, and electrical equipment.

This talk will introduce the fire hazards associated with lithium-ion batteries, electric vehicles, and BESS, explaining why these fires behave differently, what makes them challenging to control, and how engineers are responding through research, testing, and innovative design. With practical examples, we will gain an understanding of the key hazards, common misconceptions, and the contemporary approaches to managing battery fire risk.

Recent Advances in Modelling Techniques for Fire Safety and Resilience in Key Buildings



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Abstract

Fire is one of the most significant hazards affecting the built environment. Although the exact time, location, and severity of a fire cannot be predicted, its consequences can be catastrophic if appropriate fire safety measures are not implemented. Fire behavior is influenced by numerous factors, including building geometry, occupancy, fuel load, ventilation conditions, and the properties of construction materials. Consequently, each building exhibits unique fire characteristics, requiring a project-specific approach to fire safety design and assessment.

Fire safety and structural resilience are therefore fundamental considerations in the design, construction, and operation of key buildings such as high-rise buildings, hospitals, schools, airports, industrial facilities, and other critical infrastructure. The primary objective of fire safety engineering is to protect human life, followed by safeguarding property and minimizing environmental impacts. A critical component of this objective is ensuring the safe and timely evacuation of occupants during a fire emergency. This is particularly challenging in high-rise buildings, where increased building height results in longer evacuation times and greater exposure of occupants to smoke, elevated temperatures, and the potential for structural instability.

Achieving an appropriate level of fire safety requires a comprehensive assessment of fire risks and an understanding of how buildings perform under realistic fire scenarios. Performance-based fire engineering relies extensively on computational modelling, which has become one of the most effective and reliable tools for evaluating fire

performance. Recent advances in computational modelling techniques have significantly improved the accuracy and efficiency of fire safety analysis. Modern modelling tools enable engineers to simulate fire growth, smoke propagation, heat transfer, structural response, and occupant evacuation while assessing the performance of passive and active fire protection systems. These capabilities allow engineers to identify potential vulnerabilities, evaluate the effectiveness of fire protection strategies, and optimize building designs to enhance life safety, structural resilience, and overall building performance before a fire event occurs.

Research and Innovation in Fire Research from Materials to Artificial Intelligence (AI)



Prof. Kate Nguyen

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Abstract

Fire engineering is undergoing a significant transformation, driven by rapid advances in material science, digital technologies, computational modelling, and artificial intelligence (AI). As the built environment becomes increasingly complex, with the emergence of tall buildings, mass timber structures, smart infrastructure, energy storage systems, and climate-resilient designs, traditional prescriptive approaches to fire safety are no longer sufficient. Modern fire research is therefore shifting towards performance-based, data-driven, and predictive methodologies that improve both life safety and asset protection while supporting sustainable construction practices.

Recent developments in materials science have led to the introduction of innovative fire-resistant materials, advanced passive fire protection systems, engineered timber products, intumescent coatings, and composite structural systems, etc. Extensive experimental research and full-scale fire testing have enhanced the understanding of material degradation, thermal behaviour, structural resilience, and fire-induced failure mechanisms. At the same time, emerging materials used in renewable energy systems, electric vehicle batteries, and modular construction present new fire hazards that require dedicated research and updated testing methodologies.

Parallel to these developments, computational fire engineering has evolved rapidly

through advanced numerical simulations, Computational Fluid Dynamics (CFD), finite element analysis, and digital twin technologies. These tools enable engineers to accurately model fire growth, smoke movement, structural response, occupant evacuation, and firefighting interventions under realistic scenarios. Such capabilities support performance-based fire engineering by allowing designers to optimise fire safety strategies while balancing architectural flexibility, sustainability, and cost-effectiveness.

Artificial Intelligence is now emerging as one of the most transformative technologies in fire research. Machine learning algorithms, computer vision, and deep learning models are increasingly used for early fire detection, smoke recognition, intelligent alarm verification, bushfire prediction, structural damage assessment, and predictive maintenance of fire protection systems. AI-powered analysis of large datasets collected from sensors, Building Information Modelling (BIM), Internet of Things (IoT) devices, and smart building platforms enables continuous risk assessment and real-time decision-making. These technologies are also improving emergency response planning by supporting dynamic evacuation modelling, resource optimisation, and incident management.

The integration of AI with smart buildings and digital infrastructure is creating opportunities for autonomous fire safety systems capable of adapting to changing environmental conditions and occupant behaviour. However, challenges remain regarding data quality, model validation, regulatory acceptance, and the development of internationally recognised standards for AI-assisted fire engineering etc.

This presentation provides an overview of the latest research trends spanning innovative fire-resistant materials, advanced testing techniques, computational modelling, digital engineering, and artificial intelligence. Through selected international case studies and emerging applications, the session will demonstrate how multidisciplinary research is reshaping fire engineering practice and enabling safer, smarter, and more resilient built environments. The presentation will also discuss future research priorities, including AI-enabled performance-based design, sustainable fire-safe construction, automated compliance assessment, and the integration of digital technologies into next-generation fire safety engineering.

Applications and Performance of Fire-resistant Materials



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Abstract

Fire remains one of the most significant threats to the safety, functionality, and longevity of buildings and critical infrastructure. As construction technologies continue to evolve, the selection and application of fire-resistant materials have become essential in achieving compliance with modern building regulations while enhancing occupant safety and minimizing structural damage during fire events. This presentation provides an overview of the latest developments in fire-resistant materials, their performance characteristics, and practical applications across commercial, industrial, hospitality, and infrastructure sectors.

The presentation examines the role of passive fire protection systems, including intumescent coatings, cementitious fireproofing, fire-resistant boards, penetration sealing systems, fire-stopping products, and compartmentation solutions. The performance of these materials will be discussed in relation to internationally recognized testing and certification standards, including fire resistance ratings, thermal insulation, structural integrity, smoke control, and durability under varying environmental conditions.

Drawing upon regional project experience, particularly in marine and tropical environments, the presentation highlights practical considerations for material selection, installation quality, maintenance requirements, and lifecycle performance. The discussion also addresses common challenges encountered during specification and implementation, emphasizing the importance of integrating passive fire protection into the overall building design rather than treating it as a regulatory requirement alone.

Furthermore, emerging trends in sustainable fire-resistant materials, digital

specification tools, Building Information Modeling (BIM) integration, and performance-based fire engineering will be explored to demonstrate how innovation is transforming the future of fire safety.

The presentation concludes by emphasizing that effective fire protection depends not only on the performance of individual materials but also on their correct specification, certified installation, and long-term maintenance. By adopting a holistic approach to passive fire protection, stakeholders can significantly improve life safety, protect valuable assets, reduce business interruption, and enhance the resilience of modern infrastructure against fire hazards.

From Laboratory Testing to Compliance: Assessment of Passive Fire Protection Products



Dr. Imran Ahamed

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Abstract

Passive fire protection systems are critical in containing fire and smoke, maintaining structural stability, controlling fire spread and preserving escape routes during a fire event. Given they perform such a crucial role, establishing the actual contribution these systems provide requires standard fire testing; without this testing, there is no reliable data on which to base compliance decisions. This presentation provides a practical overview of how standard fire testing underpins regulatory compliance for passive fire protection products. The presentation begins with the fundamentals of standard fire resistance testing, and then outlines the key building elements commonly assessed through fire testing, including fire-rated walls, floors, fire doors, structural steel protection, service penetrations, and junction details, and explains why each requires a tailored testing approach. A central focus is the key performance criteria measured during testing – structural adequacy, integrity, and insulation – and how these relate to fire performance and can be used to meet regulatory requirements. The presentation also highlights the importance of correctly interpreting test reports, including construction details, test observations, failure modes, temperature data, field of application and limitations. Finally, the presentation discusses how laboratory test evidence is used to support compliance through product specifications, fire assessment reports, engineering judgements, and performance solutions. The key message is that fire testing provides the essential evidence base, information, and performance data that underpin overall fire compliance.

AI-Driven IoT-based fire protection for the Manufacturing Industry



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Abstract

Industrial and manufacturing facilities remain highly vulnerable to fire hazards arising from combustible dust, flammable liquids, electrical faults, and mechanical failures inherent in high-temperature, high-throughput processes. In the United States alone, an estimated 37,000 fires occur annually at industrial and manufacturing properties, causing more than one billion dollars in direct property damage each year, according to National Fire Protection Association (NFPA) analyses. On a global scale, national fire services recorded roughly 3.1 million fire incidents across 46 reporting countries in 2023 alone, while the World Health Organization estimates that burn injuries claim approximately 180,000 lives worldwide every year, disproportionately in low- and middle-income economies where manufacturing capacity is now expanding fastest. These figures underscore that conventional point-based fire detection, which relies on single smoke, heat, or ionization sensors, is frequently too slow, too spatially limited, or too prone to nuisance alarms to adequately protect large, dynamic manufacturing environments.

This work examines the design and practical application of an Artificial Intelligence (AI)-driven, Internet of Things (IoT)-based fire protection framework tailored specifically for manufacturing environments. The proposed architecture integrates three layers: (1) a sensing layer comprising modern multi-modal sensors, including photoelectric and aspirating smoke detectors, ultraviolet/infrared flame detectors, thermal-imaging cameras, gas and volatile-organic-compound sensors, and multi-criteria fusion units that combine several sensing modalities in a single device; (2) an edge-and-connectivity layer using wireless mesh networks, low-power wide-area

networks (LPWAN), and 5G to transmit sensor and video data with minimal latency; and (3) an AI and analytics layer employing convolutional neural network (CNN)-based computer-vision models, such as YOLO-family and Efficient architectures, together with predictive analytics and ensemble validation techniques, to differentiate genuine fire events from environmental noise and to enable early, condition-based response before flashover.

Drawing on documented industrial deployments and recent research benchmarks, the paper highlights measurable gains delivered by this approach: published benchmark studies report detection accuracies exceeding 80 percent with sub-20-millisecond inference times for leading object-detection architectures, while ensemble and multi-agent vision systems have demonstrated false-detection elimination rates above 90 percent, and CNN-based systems trialed across petrochemical facility camera networks have achieved fire-detection rates above 98 percent. On the global stage, the fire detection sensor market is projected to grow from approximately 2.57 billion US dollars in 2025 to 10.64 billion US dollars by 2035, a compound annual growth rate of over 15 percent that considerably outpaces the broader fire protection systems market, reflecting concentrated global investment in AI- and IoT-enabled detection technologies specifically. This growth is reinforced by tightening regulatory regimes such as NFPA 72 in North America and EN 54 in Europe, and by rising adoption across North America, Europe, and rapidly industrializing Asia-Pacific economies.

The paper further surveys practical applications across manufacturing sub-sectors, including oil and gas and petrochemical processing, automotive assembly and paint shops, electronics and semiconductor fabrication, textile and apparel mills, and warehousing and logistics, each of which places distinct demands on sensor selection and system design. Balancing these benefits, the paper also considers adoption challenges, including high upfront implementation costs, interoperability barriers arising from proprietary manufacturer protocols, expanded cybersecurity exposure from network-connected devices, and a shortage of personnel skilled in installation and maintenance. The paper concludes that AI-IoT convergence represents a structural shift in industrial fire protection, from passive, reactive alarm systems toward proactive, predictive, and increasingly autonomous safety ecosystems, offering manufacturers worldwide a scalable pathway to protect life, assets, and operational continuity.

Fire Safety Challenges and Innovative Façade Solutions in Buildings with Glass Envelopes



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Abstract

The increasing use of glass envelopes in modern buildings has significantly enhanced architectural aesthetics and energy performance; however, it has also introduced critical fire safety challenges. This presentation examines the fire behavior of glass façades, with particular focus on thermal stress-induced breakage, façade failure mechanisms, and fire spread pathways in multi-story buildings. It highlights how glass transitions from a protective barrier to a potential ventilation opening during fire, accelerating vertical fire spread and increasing risk to occupants and property.

Drawing on experimental and numerical research, the presentation introduces innovative façade solutions such as Water Wall Systems (WWS), which demonstrate improved fire resistance and delayed glass failure under high heat flux conditions. The integration of fire safety with energy performance and life cycle considerations is also discussed, emphasizing the need for a balanced, performance-based design approach. The session will provide practical insights for engineers, designers, and facilities managers on enhancing fire resilience in contemporary glass-dominated buildings.

Forensic Fire Investigation: Reconstruction of Fire Using Fire Engineering Tools



Dr. Aatif Ali Khan

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Abstract

Fire can severely damage structures and, in extreme cases, cause total building collapse. Structural and fire engineers are increasingly expected to investigate major fire-induced structural failures with the same rigor that the aviation industry applies to air crash investigations. However, conventional fire investigation methodologies — well-documented in standards such as NFPA 921, are designed primarily for small, localized compartment fires and fail to capture the unique fire dynamics of tall buildings, including travelling fire behaviour across large open floor plates, vertical spread through shafts and façades, and stack-effect-driven ventilation.

This presentation proposes a novel forensic framework to systematically reconstruct fire history in high-rise structures using available visual evidence, testimonies, and computational modelling. The methodology comprises ten sequential steps: data collection, data organisation, fire chronology mapping, timeline estimation, data refinement using witness/firefighter testimony, graphical representation on floor plans, fire progression estimation, fire scenario establishment, allocation of appropriate fire models (analytical or CFD), and structural analysis.

The framework is demonstrated through a detailed case study of the Plasco Building collapse (Tehran, 2017), a 16-storey steel-framed high-rise that collapsed following an electrical fire, killing 32 people. Using photographs, video footage, and firefighter testimonies, a ten-event timeline was established, from ignition to final collapse. The reconstructed fire history reveals travelling fire behaviour driven by high fuel loads

(textile stock), continuous open false ceilings enabling horizontal spread, and vertical spread through windows and stairwell/elevator shafts. A calibrated Fire Dynamics Simulator (FDS) model successfully reproduced both spread patterns, and subsequent heat transfer and thermomechanical analysis (via OpenSees) demonstrated highly non-uniform structural heating. Structural analysis results indicate that collapse was likely triggered by sagging floor trusses pulling in a corner column, consistent with observed collapse mechanics.

This case study demonstrates that combining forensic evidence synthesis with calibrated CFD and structural modelling produces a substantially more rigorous and realistic reconstruction of tall-building fires than idealized standard-fire curves allow. The framework offers a generalizable, evidence-driven methodology applicable to future forensic investigations of high-rise fire failures, with the broader goal of improving fire-resilient structural design and reducing life-safety risk in tall buildings.

Fire Investigation in Practice: Case Studies, Lessons Learned and Pathways to Reform



Mrs. Vithyaa Thavapalan,

Founder, Managing Director, and Forensic Fire Investigator, Forensic Origin and Cause Investigations, Australia

Abstract

Fire investigation plays a critical role in understanding how fires start, how they develop, and what can be learned to prevent future incidents. While determining the origin and cause of a fire remains central to the discipline, the findings of a fire investigation can also inform insurance decisions, legal matters, product safety, building design, risk management, emergency response, and broader fire safety reform. This presentation will explore fire investigation in practice through real-world case studies, professional lessons learned, and opportunities for improvement across the built environment. Drawing on 12 years of experience as a forensic fire investigator, the presentation will provide practical insight into the investigative process, including scene examination, evidence preservation, hypothesis development, fire dynamics, ignition source analysis, and the challenges of forming evidence-based conclusions after significant fire damage. The case studies will demonstrate the complexity of modern fire investigations across residential, commercial, industrial, vehicle, and equipment-related incidents. They will highlight the importance of considering fire patterns, witness information, electrical systems, fuel loads, ventilation, suppression activity, and post-fire disturbance before reaching an expert opinion. The presentation will also address common limitations faced by investigators, including delayed attendance, loss of evidence, incomplete information, and the risk of assumptions being made too early. A key focus will be the importance of a systematic and scientific approach to fire investigation. Conclusions must be

based on evidence, tested hypotheses, and an understanding of uncertainty. This includes recognising when a fire cause cannot be determined and why an undetermined finding may be the most appropriate and scientifically sound outcome. Beyond individual case outcomes, this presentation will consider how fire investigation can support reform by identifying recurring failures, emerging risks, and gaps in prevention. It will encourage stronger collaboration between investigators, engineers, insurers, emergency services, legal professionals, regulators, and built environment practitioners. The aim of this presentation is to show that fire investigation is not only about determining what happened, but about using the lessons from fire scenes to improve safety, strengthen resilience, support accountability, and help prevent future fires.



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